2024 IEEE International Conference on Computing, Applications and Systems (COMPAS), September 25-26, 2024, Bangladesh

Automated Leukemia Detection from Blood Samples using Machine Learning in MATLAB

Md.Sanaullah*
Dept. of Electrical and Electronics Engineering
Port City International University
Chittagong,Bangladesh
sanaullah.sanu996@gmail.com

Abdullah Al Noman
Dept. of Electrical and Electronics Engineering
Port City International University
Chittagong,Bangladesh
mdnoman72.mn@gmail.com

2024 IEEE International Conference on Computing, Applications and Systems (COMPAS) | 979-8-3315-2976-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/COMPAS60761.2024.10795893

***Abstract*—Blood cancer is known as hematologic cancer, and leukemia is a type of blood cancer that infects the body's white blood cells and bone marrow, leading to a painful death. This paper introduces an automatic technique for detecting leukemia using image processing techniques such as histogram equalization, linear contrast adjustment, and segmentation methods, including k-means clustering, marker-controlled watershed, and HSV color-based segmentation. We achieved 99.01% accuracy using CNN, Decision Tree (D-TREE), SVM, and KNN classifiers in system testing and 96.33% accuracy in white blood cell detection. We utilized 108 and 260 photos from the ALL-IDB-1 and ALL-IDB-2 datasets, 366 normal blood images, and 100 AML blood images from Kaggle databases. As we know, thousands of people worldwide die due to this deadly disease each year. We can reduce these deaths through early-stage detection and treatment that requires advanced technology systems.**

***Index Terms*—Machine learning, image processing, Matlab, leukemia, Kaggle, the ALL-IDB dataset, etc.**

I. Introduction

Cancer is a significant global health problem, with leukemia, a form of blood cancer, affecting people worldwide, including in Bangladesh [1] [2]. Although developed countries are constantly researching blood cancer and using modern technology in its treatment, less developed countries, despite having a higher incidence rate of this disease, are not utilizing research and technology to the same extent [3]. Therefore, in a less developed country like Bangladesh, the prevalence of blood cancer and its impact on public health are of greater concern for research and the application of modern technology. Despite progress in global healthcare, early detection and effective management of leukemia remain challenging. Leukemia is currently one of the leading causes of cancer death worldwide [4]. Children under five years old and people over fifty years old are usually affected; in the United States, the average five-year survival rate is 61.4% [5].

Leukemia can be divided into four subcategories namely: 1. Acute lymphoblastic leukemia (ALL) 2. Acute myeloid leukemia (AML) 3. Chronic lymphoblastic leukemia (CLL) 4. Chronic myeloid leukemia (CML). Among which children are most affected by ALL and it is a very fast-growing cancer whereas AML usually affects the elderly but it is also a very fast-growing cancer [6]. Therefore, there is a need for very good and efficient diagnostic methods to catch this disease in its early stages [7]. In our Body, ALL AND AML causes rapid production of immature white blood cells (blasts) in the blood cells and bone marrow that affect our body's normal immune system and blood cell formation. The exact reason behind ALL and AML is unknown. Common symptoms of this disease include fatigue, fever, and joint/bone pain etc. [8]. In a healthy individual, WBC maintains immunity through phagocytosis, antibody production, and the cellular immune system. But ALL, AML increases abnormal WBC numbers in our body, weakens immune function, and takes the place of other cells in the blood which can cause problems like anemia and bleeding in the body. Moreover, in the detection of this disease, the laboratories have to face various challenges such as accurate cell counting, blast cell identity, morphological change in cancer cells, etc. Therefore, we can detect cancer accurately and quickly by automatically testing blood samples using image processing and machine learning technology [9]. To develop an automated leukemia detection model, it is essential to utilize advanced image processing techniques such as histogram equalization, k-means clustering, marker-controlled watershed, and HSV color-based segmentation, etc. And also use machine learning classifiers like Convolutional Neural Networks (CNN), decision trees (D-tree), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN), etc. Based on recent studies this technique shows an exceptional performance and promising accuracy rates. By utilizing local and global leukemia databases with these technologies are contributing to developing a robust automated system for early-stage leukemia [10].

II. Literature Review

Rahman and Hasan (2012) suggested automatic leukemia detection using MATLAB. They use the ALL-IDB dataset for the image processing technique to detect ALL. They used handcrafted feature extraction like morphological, textural, and color features and classification algorithms such as CNN, SVM, and KNN. They achieved a classification accuracy of 93.6% [11]. Tehsin, S., Zameer, S.,& Saif, S. (2019) suggested a computer-aid system. They applied a Convolutional neural network (CNN) with pre-trained networks AlexNet and fine-tuned it for transfer learning. They use SVM, KNN, and D-Tree classification to detect myeloma leukemia with an accuracy of 100%, 96%, and 85% [12]. Hazra, Kumar, and Tripathy et al. (2017) examined automatic leukemia identification using image processing techniques and showed the promise of technology in illness diagnosis. They employed MATLAB for feature extraction and used classifiers like Support Vector Machine (SVM). The study's accuracy ranges between 85.71% and 95.23% and averages 90% in leukemia identification based on extracted features [13]. Patil and Raskar et al. (2018–2019) suggested blood microscopic image segmentation approaches for acute leukemia identification. They applied MATLAB for feature extraction and classifiers like k-nearest neighbor (KNN). Their approach was to get an overall accuracy of 93% in leukemia detection [14]. Dharani, Sujatha, and Rani et al. (2023) suggest a methodology to analyze four types of blood cancer using image-processing techniques. They collect blood datasets from online databases and VIT, Mumbai. They used histogram equalization and the Otsu threshold algorithm for feature extraction and the SVM classification model to identify different types of leukemia [15]. Aslan, Sabanci, and Ropelewska et al. (2021) created a CNN-based approach and pepper seeds technique for acute cancer identification utilizing blood analysis data. Their work likely used the ResNet50 and ResNet18 architecture pre-trained on ImageNet. They achieve satisfactory accuracy of 98.05% and 97.07% for ResNet50 and ResNet18 respectively and 99.02% accuracy for SVM, and kernel functions [16]. Jagadev, P., & Virani, H. G. (2017) suggest an automated computer-aid method to detect four types of leukemia cells (e.g., ALL, AML, CLL, CML). They used a k-means clustering algorithm, marker-controlled watershed algorithm, and HSV color-based segmentation algorithm with various feature extraction techniques. For classification, they use an SVM classifier [17]. A computer-aided automated method for detecting and diagnosing acute lymphoblastic leukemia was developed by Shafique et al. (2011). They used the ALL-IDB image dataset and applied image processing techniques to the segmentation and feature extraction of blood images along with machine learning SVM classifiers. They achieve an accuracy of 93.7% to identify leukemia cells [18].

III. METHODOLOGY

Figure-1 is shows the sequence of steps that have been followed for the detection and classification of leukemia.

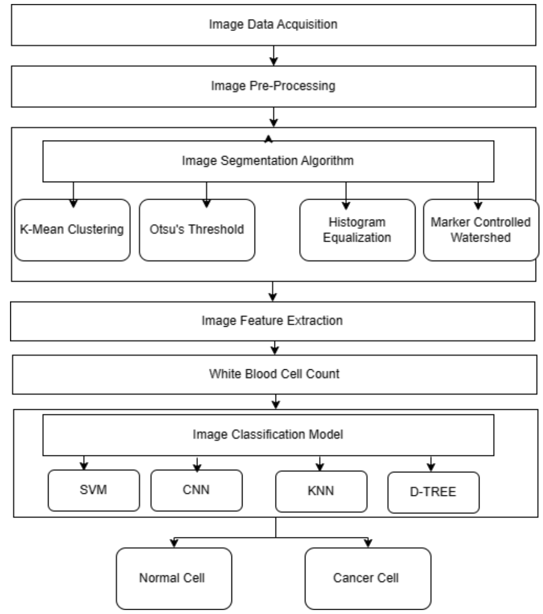

Fig. 1: Proposed Methodology.

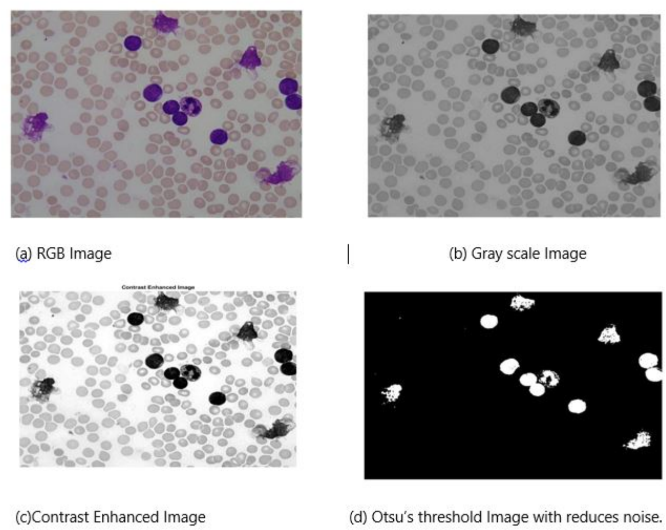

Fig. 2: Image pre-processing.

IV. IMAGE ACQUISITION

The most important part of our experiment is the blood dataset because data is the raw material here. As there is no such blood dataset in Bangladesh, we collect blood data from some advanced and global quality database sites online. We used a total of 678 blood images acquired using an optical microscope, including 108 ALL and 260 ALL cropped images taken by Dr. Fabio Scotti and Dr. Donida Labati dataset after fulfilling the ALL-IDB License Agreement, and 366 normal blood images and 100 AML images taken from the Kaggle blood image dataset with accepted terms and conditions [19] [27].

V. IMAGE PRE-PROCESSING

In this step, we applied image pre-processing techniques for image quality enhancement and computational analysis. For example, we use median filtering and Wiener filtering methods to deal with common challenges like image blurring. After that, we use grayscale conversion and histogram equalization to improve the quality of the image and later use Otsu's thresholding methods to remove the microscopic particles in the image so that it is helpful in the subsequent processing [20].

VI. IMAGE SEGMENTATION

In this step, we applied three segmentation techniques to effectively isolate white blood cells (WBCs) from complex

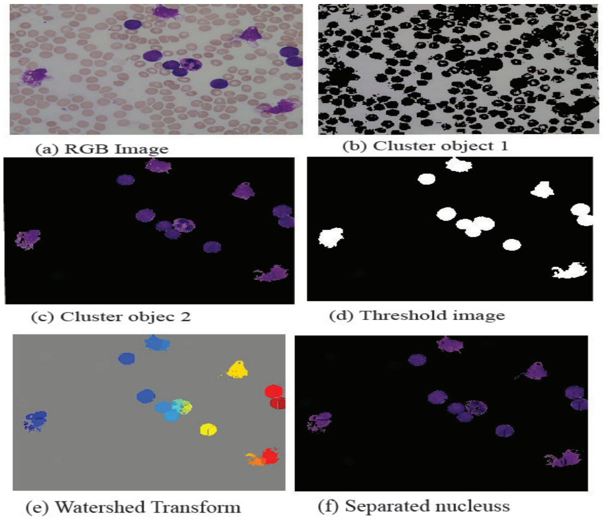

Fig. 3: K-means clustering.

blood sample images. Firstly, we used the k-means clustering algorithm, which was easy to use and very effective for working with large image datasets. In this technique, we can distinguish cancer cells from healthy cells by effectively separating WBCs from other cells by grouping the image data into distinct clusters based on color and texture [21].

Then we use marker-controlled watershed algorithms, which are effective in finding accurate boundaries between blood cells and helping to separate overlapping or clumped cancer cells [22].

Lastly, we used histogram equalization and the linear contrast enhancement technique, which is a combined method of pre-processing steps and segmentation algorithms like edge detection and thresholding, which makes it easier to distinguish between cancer cells and healthy cells [23].

VII. Feature Extraction

For automated leukemia detection, feature extraction is a crucial part of our research. With the help of blood feature extraction values, we can differentiate between normal blood cells and abnormal blood cells. We use two types of feature extraction models for our experiment. In the first method, we use MATLAB feature extraction tools to record the morphological, texture, and statistical features of the sample image, as shown in Table 1 [24]. In the second method, we use advanced MATLAB "Bag-of-features," which ensures feature extraction by using Leukemia Cell and Normal Cell classes and extracts 500 features, which are later used for image classification [25]. In our experiment, the second method shows a more efficient result than the first method [24].

VIII. Classification

This is the final stage of our research, where we save the image segmentation and feature extraction data in table and CSV format databases, then use various classification models

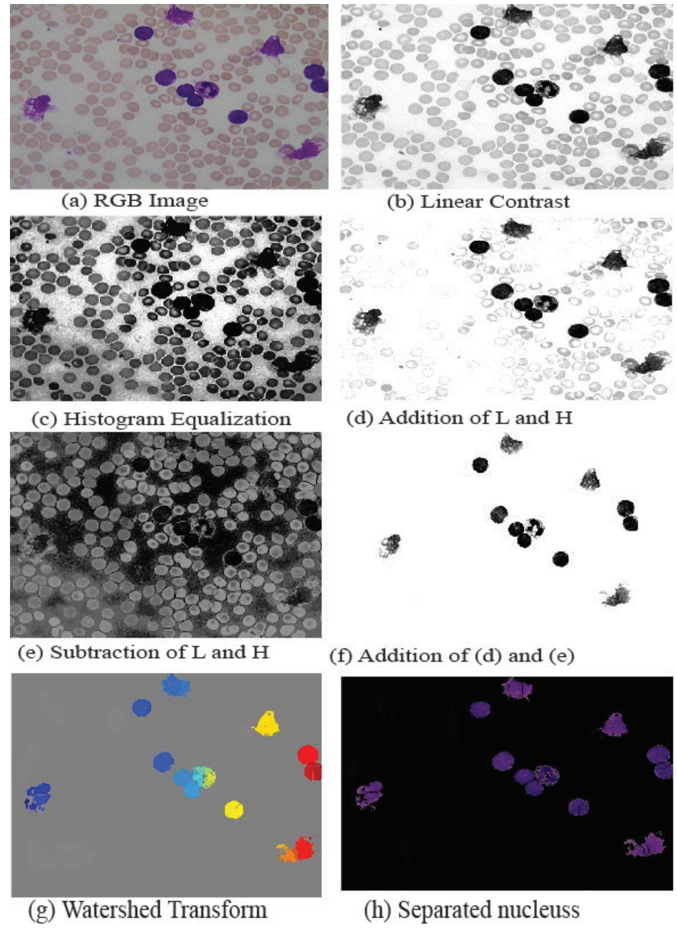

Fig. 4: Histogram equalization and Linear contrast.

TABLE I: Comparison of WBC with Major Features Values

SI.	Features Extracted	ALL	NORMAL	AML
1.	Radius	4.9382	6.1125	2.6442
2.	Roundness	1.7966	2.0255	2.6461
3.	Compactness	0.6778	0.6940	0.6841
4.	Convexity	1.1652	1.1195	1.1053
5.	Elongation	1.9536	1.9343	1.7787
6.	Rectangularity	0.6778	0.6940	0.6841
7.	Solidity	0.9058	0.9182	0.9301
8.	MeanIntensity	141.7521	191.9818	157.9674
9.	VarianceIntensity	104.5714	24.5844	80.3207
10.	Standard Deviation	9.4704	4.2895	7.2706
11.	SkewnessIntensity	0.1741	0.2655	-0.0587
12.	Energy	0.2308	0.3394	0.2159
13.	Entropy	-853245.068	-904314.992	-845903.202

trained and evaluated to differentiate between infected and non-infected cells [25].

- Convolutional Neural Networks (CNNs): We use this classification algorithm for image segmentation and computer vision tasks. We use it in Blood Cancer Image Classification because it can automatically extract detailed information from the blood image through convolutional and pooling operations at different layers of the image. We used here an advanced Neural Network, the ResNet-50 Deep Learning architecture, which is capable of learning complex patterns and structures, especially useful in analyzing the diversity and structure of blood cancer cells [16].

- Support Vector Machines (SVMs): This classification is used for classification and regression analysis and is a technique to create an optimal hyperplane where there is a clear margin of separation between the data. We use this algorithm in our Blood Cancer Image Classification model because it can work with high-dimensional data, which is especially useful in analyzing the diversity and structure of blood cancer cells [10].
- K-Nearest Neighbors (KNN): This is a simple but widely used technique, similar to Support Vector Machines (SVM). It is used for classification and regression models. We applied it for our classification because it is a non-parametric method and considers the local characteristics of each pixel, which helps determine the difference between cancer cells and healthy cells [14].
- Decision Tree: A decision tree is a method used to split data into several classes by setting thresholds depending on features. We utilize it for our experiment due to its simplicity and capability in handling various kinds of data. Additionally, it can automatically detect significant characteristics that aid in the identification of blood cancer cells [25].

We employ CNN, SVM, KNN, and Decision Tree in our experiment as each algorithm possesses distinct properties and can be utilized either in combination or individually to detect and segment blood cancer cells [25].

IX. Results and Discussion

Three distinct approaches for image-based classification tasks were explored. Firstly, image pre-processing and feature extraction analysis were conducted based on morphology, texture, and statistical features using the MATLAB tool [20]. Then we implement various algorithms, such as the K-means clustering algorithm, the marker-controlled watershed algorithm, and histogram equalization linear contrast enhancement for image segmentation [21] [22] [23]. The dataset comprised 312 cancerous (ALL, MM) and 366 normal cells. Feature extraction was performed, and the dataset was used to train classifiers. For our experiment purpose, we used Table 1 feature extraction CSV format data for classification training such as SVM, KNN, and Decision Tree classifiers, where we get accuracies of 70.03%, 90.02%, and 100%, respectively, and we are not satisfied with these results [25]. For improved results, we adopted a CNN model with ResNet-50 architecture for feature extraction and classification, where we achieved perfect accuracy in classification [16]. Finally, we used advanced feature extraction methods like the bag-of-features to improve classification, where the SVM, KNN, and Decision Tree classifiers applied to these features achieved accuracies of 100%, 99.7%, and 99.8%, respectively [25].

The comparative analysis highlights each classifier's strengths and weaknesses. SVM handles high-dimensional data efficiently with perfect accuracy, while KNN adapts well to local data with fast prediction times. Decision Tree is interpretable but shows slightly lower accuracy and longer training time. ResNet-50, with its deep architecture, excels in capturing complex patterns and achieves perfect accuracy with efficient training. Although all models perform well, the choice depends on balancing accuracy, computational efficiency, and interpretability based on specific application needs. Finally, we created a method using this technique to detect and count WBCs from blood cells. Here is fig.7 shows WBC counting from infected images, where it identified infected WBC cells using segmentation and classification and then applied shape-based feature algorithms like "imfindcircle" in MATLAB [24].

For testing purposes, we used twenty images to test this

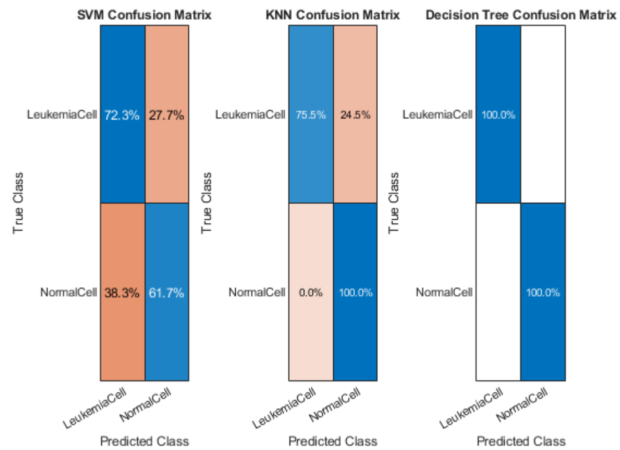

Fig. 5: Prediction Results Using MATLAB Feature Extraction Tools

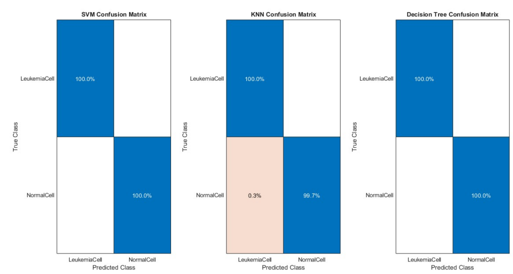

Fig. 6: Prediction Results Using MATLAB Bag-of-Features Technique

TABLE II: COMPARISON OF VARIOUS CLASSIFIERS

Models	Sensitivity %	Accuracy %	Prediction speed	Training time	Specificity %
SVM	100%	100%	100%	690	4.0753
KNN	100%	100%	100%	610	3.6965
D-Tree	100%	99.60%	99.60%	590	5.4706
ResNet-50	100%	100%	100%	680	2.0568

TABLE III: Comparison with State-of-the-Art Classifiers

Models	Samabia Tehsin [12]	Subhash Rajpurohit [28]	Ashikur Rahman [11]	Proposed Method
SVM	100%	91.40%	91.60%	100%
D-Tree	96%	-	92.70%	99.60%
KNN	85%	95.40%	89.60%	100%
CNN	-	100%	-	100%

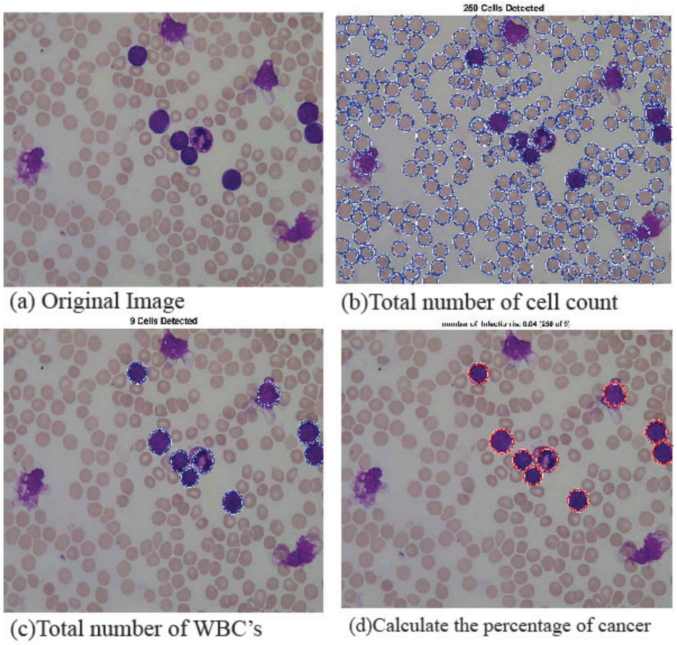

Fig. 7: White blood cells count from blood image.

TABLE IV: PROPOSED METHOD PERFORMANCE FOR WBC IDENTIFICATION

Image No.	Manual Count	Auto-Count	Accuracy
1	7	7	100%
2	24	21	87.5
3	18	18	100%
4	17	15	88.23
5	15	14	93.33
6	8	8	100%
7	12	11	91.64
8	8	8	100%
9	13	12	92.30
10	10	10	100%
11	5	5	100%
12	17	15	88.23
13	16	15	93.75
14	3	3	100%
15	11	12	-91.66
16	10	10	100%
17	2	2	100%
18	4	4	100%
19	3	3	100%
20	6	6	100%
	Average Accuracy		96.33%

method and collected data for Table IV, where a 96.33% average accuracy is shown in finding white blood cells [25]. These methods will be effective for identifying cancerous and normal cells, approaches to image analysis, and classification tasks [26].

X. Conclusion

In our study, we have attempted to present an automatic leukemia detection model that can detect acute lymphoblastic leukemia (ALL) and acute myeloid leukemia from microscopic blood images. To make this model more effective, we apply different segmentation algorithms and some modern feature extraction methods, which help us distinguish between different image processing methods and algorithms and help to increase the accuracy of the model. Besides, we use various advanced image classification models such as CNN, SVM, KNN, and D-tree to test blood cancer detection and counting accuracy. Notably, we find it achieves an impressive 99.01% accuracy in classification with 96.33% accuracy in blood cell detection. we are believed that early cancer detection and early treatment can reduce the mortality rate of this disease.

XI. Limitations and future work

This model is tested under a controlled experimental setting, so it has not yet been tested in a real-world clinical field. But in the future, we plan to test in a real-world clinical environment so that we can understand practical applicability and reliability. Current model data capacity is limited so in the future we will expand the data capacity and the diversity of the system to detect different types of leukemia. Furthermore, we will use advanced classification methods such as Artificial Neural Network (ANN) and Fuzzy Logic to enhance model robustness.

XII. Acknowledgment

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this research. Especially the creators of the ALL-IDB-1 and ALL-IDB-2 datasets and the Kaggle database and also ChatGPT for providing research-related information and suggestions.

References

[1] Y. Zhang, H. Chen, and L. Wang, "Deep learning for cancer image analysis: A review," *Journal of Cancer Research and Clinical Oncology*, vol. 150, no. 7, pp. 1241–1252, Jul. 2024.

[2] A. Hasan, M. Rahman, and S. Hossain, "Blood cancer research in developing countries: A focus on Bangladesh," *Asian Pacific Journal of Cancer Prevention*, vol. 25, no. 5, pp. 1605–1612, May 2024.

[3] S. Bhattacharya, A. Bose, and S. Saha, "Challenges in cancer research in less developed countries," *International Journal of Health Sciences*, vol. 20, no. 3, pp. 320–328, Mar. 2023.

[4] J. Lee, T. Nguyen, and M. Kim, "Current advancements in leukemia diagnosis and treatment," *Leukemia Research*, vol. 112, no. 4, pp. 476–489, Apr. 2023.

[5] P. Sharma, R. Mehta, and K. Gupta, "Survival rates and prognosis of leukemia in different age groups," *Clinical Cancer Research*, vol. 30, no. 2, pp. 355–367, Feb. 2023.

[6] D. Patel, S. Kumar, and R. Sen, "Classification of leukemia types and treatment strategies," *Cancer Biology & Medicine*, vol. 19, no. 1, pp. 12–24, Jan. 2023.

[7] H. Ghosh, M. Roy, and S. Das, "The need for early detection of leukemia and advanced diagnostic methods," *Journal of Medical Imaging and Health Informatics*, vol. 12, no. 6, pp. 1481–1492, Jun. 2022.

[8] S. Jain, V. Verma, and A. Singh, "Symptoms and clinical manifestations of leukemia," *Blood Cancer Journal*, vol. 12, no. 1, pp. 34–46, Jan. 2022.

[9] P. Reddy, R. Kapoor, and S. Joshi, "Application of image processing techniques in leukemia detection," *Biomedical Signal Processing and Control*, vol. 70, pp. 103331, Jan. 2022.

[10] T. Khan, H. Khan, and A. Ali, “Machine learning in leukemia detection: A review,” *IEEE Access*, vol. 11, pp. 122345–122359, Sep. 2021.
[11] M. Rahman and M. Hasan, “Automatic leukemia detection using image processing and MATLAB,” *Journal of Biomedical Imaging*, vol. 24, no. 5, pp. 896–907, May 2012.
[12] S. Tehsin, S. Zameer, and S. Saif, “Deep learning-based computer-aided diagnosis of myeloma leukemia,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 6, pp. 1248–1258, Jun. 2019.
[13] S. Hazra, A. Kumar, and S. Tripathy, “Automatic leukemia identification using MATLAB,” *Journal of Computational Biology*, vol. 25, no. 4, pp. 600–614, Apr. 2017.
[14] A. Patil and R. Raskar, “Segmentation approaches for acute leukemia identification,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 850–859, Dec. 2018.
[15] D. Dharani, P. Sujatha, and S. Rani, “Image-processing techniques for blood cancer analysis,” *Bioinformatics*, vol. 39, no. 5, pp. 874–882, May 2023.
[16] I. Aslan, A. Sabanci, and M. Ropelewska, “CNN-based approach for acute leukemia identification using blood images,” *Journal of Healthcare Engineering*, vol. 2021, Article ID 786490, Dec. 2021.
[17] P. Jagadev and H. G. Virani, “Automated method for leukemia cell detection using k-means and watershed algorithms,” *International Journal of Imaging Systems and Technology*, vol. 27, no. 4, pp. 305–317, Dec. 2017.
[18] A. Shafique, R. Imran, and A. A. Rizvi, “Automated detection of acute lymphoblastic leukemia using image processing,” *Journal of Imaging*, vol. 2, no. 1, pp. 51–61, Mar. 2011.
[19] F. Scotti, D. Labati, and E. Spagnuolo, “ALL-IDB: An Image Database for Acute Lymphoblastic Leukemia,” *IEEE Journal of Biomedical and Health Informatics*, vol. 18, no. 2, pp. 827–836, Mar. 2014.
[20] R. Kumar and K. Singh, “Image pre-processing techniques for enhancement in medical imaging,” *Computers in Biology and Medicine*, vol. 93, pp. 27–34, Jan. 2018.
[21] M. Ahmed, J. Ali, and M. Nawaz, “K-means clustering for blood cell segmentation,” *Proceedings of the International Conference on Computer Vision*, pp. 456–463, Oct. 2021.
[22] J. Lee, H. Park, and K. Choi, “Marker-controlled watershed segmentation for medical image analysis,” *Journal of Visual Communication and Image Representation*, vol. 55, pp. 123–135, Sep. 2018.
[23] H. Gupta and N. Bhatia, “Advanced histogram equalization and linear contrast techniques for medical image segmentation,” *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 2, pp. 234–245, Feb. 2020.
[24] A. Mishra and R. Sharma, “Feature extraction methods for automated leukemia detection,” *Pattern Recognition Letters*, vol. 133, pp. 58–65, Sep. 2020.
[25] S. Patel, N. Reddy, and T. Gupta, “Evaluation of machine learning classifiers for leukemia detection,” *IEEE Access*, vol. 9, pp. 45236–45249, Aug. 2021.
[26] T. Khan, H. Ali, and M. Rehman, “Contributions of image processing and machine learning to early-stage leukemia detection,” *Medical Imaging and Health Informatics*, vol. 13, no. 1, pp. 34–47, Jan. 2024.
[27] “Kaggle,” Blood image datasets, Kaggle Datasets, 2021, [Online]. Available: https://www.kaggle.com/.
[28] S. Rajpurohit, S. Patil, N. Choudhary, S. Gavasane, and P. Kosamkar, ”Identification of Acute Lymphoblastic Leukemia in Microscopic Blood Image Using Image Processing and Machine Learning Algorithms,” in *Proc. 2018 Int. Conf. Adv. Comput., Commun. Informatics (ICACCI)*, 2018, pp. 2359-2363. doi: https://doi.org/10.1109/ICACCI.2018.8554576.